

# California Fault Model Explores a Changing 'Earthquake Gate'

From: ScienceDaily All Mined: 2026-10-03 5 min read

Earthquake faults do not always act independently. At Cajon Pass, northeast of Los Angeles, the San Andreas and San Jacinto fault systems come close together. A new study describes this **junction** as an earthquake gate whose behavior changes as stress builds and shifts.

A fault is a break in Earth's crust along which rock can move. Sections may become stuck while forces continue pushing them. This allows stress to **accumulate** until movement releases stored energy in an earthquake. A **rupture** is the breaking and movement that spreads along a fault during that event.

An international team led by Dr. Liliane Burkhard of the University of Bern modeled roughly 1,000 years of earthquakes along the two Southern California fault systems. The study was published in the Journal of Geophysical Research: Solid Earth.

Its results suggest that stress in parts of the system has reached, and in some places exceeded, the highest values found in the model's 1,000-year history. These findings are **estimates** produced by a **simulation**.

The researchers built a physics-based, four-dimensional earthquake cycle model. Here, four dimensions means three dimensions of space plus time. The model tracks how earthquakes change stress on neighboring fault sections, how stress **accumulates** during quiet periods, and how deeper crustal layers gradually relax after large **ruptures**.

To reconstruct past activity, the team incorporated evidence from **radiocarbon** dating, unusual tree-ring patterns and historical accounts of **ruptures** that broke the ground surface. Combining those records with physics allowed them to explore changes across centuries.

Cajon Pass was a central focus because a **rupture** reaching the **junction** can behave in different ways. During the magnitude 7.9 Fort Tejon earthquake in 1857, the **rupture** stopped there rather than continuing onto the San Jacinto Fault. The Wrightwood earthquake in 1812 behaved differently, passing through the **junction** and rupturing both systems.

Burkhard explains that Cajon Pass responds to stress conditions, which change over centuries. The gate description therefore does not mean there is a fixed barrier underground. Whether a **rupture** stops or continues depends on stress conditions around the **junction**.

The broader Los Angeles region has not experienced a comparable major earthquake since 1857, according to the source, while stress has continued building along parts of the system. This study helps researchers investigate how connected faults respond to that long-term buildup.

## Word Treasure

**junction** -- A place where different parts meet or connect.

**accumulate** -- To build up over time.

**rupture** -- Breaking and movement spreading along a fault during an earthquake.

**simulation** -- A computer representation used to explore how a system behaves.

**radiocarbon** -- A radioactive form of carbon used to estimate the age of once-living material.

**estimates** -- Approximate values worked out using available information.

## Background you need

A rupture is the breaking and movement that spreads along a fault during an earthquake. The model follows changes through three dimensions of space and through time. The study's stress findings are estimates from a simulation.

## Break it down

Introduces the connected faults and Cajon Pass.  
Defines faults, stress buildup and ruptures.  
Identifies the research team and publication.  
Presents the model's stress estimates.  
Explains the model and the historical evidence.  
Compares the Fort Tejon and Wrightwood ruptures.  
Attributes the changing gate explanation to Burkhard.  
Connects the study to long-term stress buildup.

## Why it matters

The study helps researchers understand how stress changes across connected Southern California faults and how those conditions may affect a rupture at Cajon Pass.

## Test what you remember

**1. Who led the international research team?**

- ☐ (A) Dr. Liliane Burkhard of the University of Hawai'i
- ☐ (B) Dr. Liliane Burkhard of the University of Bern
- ☐ (C) Dr. Liliane Burkhard of the U.S. Geological Survey
- ☐ (D) Dr. Liliane Burkhard of UC San Diego

**2. What does four-dimensional mean in this study's model?**

- ☐ (A) Four separate dimensions of space
- ☐ (B) Four separate periods of earthquake history
- ☐ (C) Two dimensions of space plus two faults
- ☐ (D) Three dimensions of space plus time

**3. Which set of evidence helped reconstruct past earthquake activity?**

- ☐ (A) Radiocarbon dating, tree-ring patterns and historical accounts
- ☐ (B) Satellite photographs, traffic records and population surveys
- ☐ (C) Ocean temperatures, rainfall records and wind measurements
- ☐ (D) Soil temperatures, river levels and weather forecasts

**4. What do the model's results suggest about stress in parts of the system?**

- ☐ (A) It disappears completely after each modeled earthquake.
- ☐ (B) It remains unchanged throughout the model's history.
- ☐ (C) It matches or exceeds the model's historical highs.
- ☐ (D) It stays below the model's lowest historical values.

**5. How did the two historical ruptures behave at Cajon Pass?**

- ☐ (A) Fort Tejon stopped; Wrightwood passed through.
- ☐ (B) Fort Tejon passed through; Wrightwood stopped.
- ☐ (C) Both Fort Tejon and Wrightwood stopped.
- ☐ (D) Both Fort Tejon and Wrightwood passed through.

**6. According to Burkhard, what affects Cajon Pass's behavior?**

- ☐ (A) A fixed barrier that never changes
- ☐ (B) The names given to the fault systems
- ☐ (C) Stress conditions that change over centuries
- ☐ (D) The number of researchers studying it

Answer key (try the quiz first!): 1: B 2: D 3: A 4: C 5: A 6: C

## Questions to think about

**1. Dr. Liliane Burkhard:** Burkhard explains that Cajon Pass responds to stress conditions that change over centuries, so the junction can stop a rupture or allow it to continue.

**MY THOUGHTS (at least 20 words):**